Bayesian Optimization of Multi-Material 3D-Printed Tensegrity Structures for Energy Absorption

BYU Ira A. Fulton College of Engineering
Mentored Research Grant Proposal

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Co-PI: Sterling G. Baird, Department of Mechanical Engineering

Duration: 2 Years — **Requested:** $25,000

Abstract

We propose a two-year mentored research program in which 2–3 undergraduate students design and optimize multi-material 3D-printed tensegrity-inspired structures for energy absorption. Tensegrity-inspired structures—assemblies of rigid compression members (PLA) and flexible tension members (TPU)—offer tunable mechanical responses and high strength-to-weight ratios. Using a multifidelity Bayesian optimization framework, students will combine finite element simulations (low-fidelity) with physical experiments (high-fidelity) to efficiently navigate the high-dimensional design space. Students will gain hands-on experience in CAD design, multi-material 3D printing, impact testing, computational modeling, and data-driven optimization. Expected outcomes include undergraduate conference presentations (UCUR, ASME IDETC), a peer-reviewed journal submission, and development of marketable technical skills. This project provides a rich mentored learning environment at the intersection of structural mechanics, advanced manufacturing, and machine learning.

Number of undergraduates: 2–3
Number of graduate students: 1 (co-mentor)

Budget Summary

Category	Year 1	Year 2	Total
Undergraduate wages	$7,500	$7,500	$15,000
Graduate student wages	$2,000	$2,000	$4,000
Supplies	$2,750	$750	$3,500
Travel	$0	$2,000	$2,000
Other	$250	$250	$500
Total	**$12,500**	**$12,500**	**$25,000**

Relationship to External Funding

The PIs do not currently hold external funding for tensegrity or architected materials research. This MRG provides the primary support for establishing a mentored undergraduate research program in this area. If the program proves successful, the preliminary data and mentoring framework developed here may inform a future NSF proposal; however, the MRG is self-contained and its outcomes—trained student researchers, conference presentations, and a journal submission—do not depend on external funding.

1 Research Motivation and Overview

We propose a two-year mentored research program in which 2–3 undergraduate students design and optimize **multi-material 3D-printed tensegrity-inspired structures** for energy absorption. Tensegrity (tensional integrity) structures—assemblies of rigid compression members suspended within a continuous network of tension members—exhibit remarkable strength-to-weight ratios and tunable mechanical responses, making them attractive for protective equipment, packaging, and aerospace applications [Skelton and de Oliveira, 2009, Fraternali et al., 2015]. Recent work has shown that *tensegrity-inspired* architectures can be directly 3D-printed and retain desirable load-limiting behavior under impact [Pajunen et al., 2019]. Because multimaterial FDM TPU elements are not ideal cables, we target tensegrity-inspired architectures that reproduce key tensegrity-like mechanical responses while remaining manufacturable; notably, TPU's rate-dependent damping is an *asset* for energy absorption rather than merely a deviation from idealized behavior.

Recent advances in multi-material 3D printing enable the fabrication of tensegrity-inspired structures on a single print platform, using PLA for rigid struts and TPU for flexible tension elements [Ye et al., 2023, Khatri and Egan, 2024]. This eliminates tedious manual assembly and opens a rich, high-dimensional design space. Navigating this space efficiently requires intelligent experimental design; **multifidelity Bayesian optimization** (BO) provides exactly this capability by combining inexpensive simulations (low-fidelity) with physical experiments (high-fidelity) to guide the search toward optimal configurations [Mo et al., 2023, Perdikaris et al., 2017].

The primary objective of this project is to provide a rich, interdisciplinary mentored research experience for undergraduates. Students will gain hands-on ~~experience in CADdesign~~skills in CAD, multi-material 3D printing, impact testing, computational modeling, and data-driven ~~optimization—skills directly transferable to graduate school and engineering careers~~optimization. Each student will lead a ~~clearly~~ scoped research project, present at conferences, and ~~contribute to a peer-reviewed~~ co-author a publication.

~~**Figure placeholder:** Overview of project approach—tensegrity unit cell design (left), multi-material 3D printing of PLA struts and TPU tension elements (center), and multifidelity Bayesian optimization loop connecting simulations to experiments (right).~~ **Figure placeholder:** Ten
printing with wra

Figure 1: Conceptual overview of the proposed research and mentoring framework.

2 Background

Tensegrity structures consist of isolated compression members (~~bars or~~ struts) held in ~~stable~~ equilibrium by a continuous network of tension members ~~(cables or strings)~~ [Skelton and de Oliveira, 2009, Sultan, 2009]. Their defining ~~characteristic—that no~~ characteristic—no two rigid bars touch—yields lightweight assemblies with surprising load-bearing capacity ~~, resilience,~~ and tunable nonlinear responses [Amendola et al., 2014, Zhang et al., 2015]. Pajunen et al. [Pajunen et al., 2019] demonstrated that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact~~, confirming their potential for energy absorption without requiring manual cable assembly. Prior work on multi-material~~. Multi-material rigid–soft printing [Ye et al., 2023, Khatri and Egan, 2024] has shown that PLA–TPU combinations printed on a single FDM platform ~~can~~ achieve tunable energy absorption~~, though interface durability under repeated impact remains an active area of investigation~~. Ye et al. demonstrated a wrapping-based strategy (rigid cores encapsulated by continuous soft skins) that prevents interface delamination and enables cyclic durability—a critical enabler for our fabrication approach.

Bayesian optimization (BO) is a sequential, model-based strategy for optimizing ~~expensive-to-evaluate~~ expensive black-box functions ~~[Shahriari et al., 2016, Frazier, 2018]. A~~ using a Gaussian process surrogate ~~model is fitted to observed data, and an acquisition function balances exploration with exploitation~~[Shahriari et al., 2016, Frazier, 2018]. BO has been applied to materials discovery [Lookman et al., 2019] and metamaterial design [Wang et al., 2022]. ~~Critically,~~ Mo et al. [Mo et al., 2023] ~~recently~~ demonstrated **multifidelity BO** for architected materials, using ~~inexpensive~~ simulations as a low-fidelity source and ~~physical~~ experiments as high-fidelity data. We adopt this framework~~directly: finite element~~: FEA simulations serve as the low-fidelity source, and 3D-printed tensegrity impact tests provide high-fidelity validation. ~~An early calibration campaign will verify low-fidelity–high-fidelity correlation before full optimization begins, since~~ Because MFBO effectiveness depends on adequate correlation between fidelity levels, we establish a Year 1 Go/No-Go decision gate: students will validate simulation–experiment correlation on a small calibration set before launching full optimization.

3 Student Research Project 1: Simulation and Bayesian Optimization

Student profile: One junior or senior undergraduate with coursework in mechanics of materials or computational methods.

Scope and Deliverables

This student will develop and run the computational pipeline:

1. **Parameterize a tensegrity unit cell** suitable for simulation and multi-material 3D printing, including strut geometry (PLA), tension element geometry (TPU), topology, and boundary conditions.
2. **Build finite element models** using ABAQUS or open-source FEA tools (e.g., FEniCS) to predict force–displacement response and energy absorption under compaction and drop-test loading. These simulations serve as the *low-fidelity* source in the multifidelity framework. ~~An initial calibration set will validate simulation–experiment correlation before full optimization begins~~
3. **Validate LF–HF correlation (Go/No-Go gate):** Before launching full optimization, the student will run a small calibration set comparing simulation predictions against experimental results on single unit cells. If the correlation (R^2) between static compliance and dynamic peak force is < 0.8, the team will pivot to single-fidelity experimental BO, ensuring feasibility regardless of model accuracy.
4. **Implement a multifidelity Bayesian optimization loop** (following Mo et al. [Mo et al., 2023]) using Python libraries (BoTorch/Ax) to efficiently explore the design space, fusing simulation predictions with experimental results from Student 2.
5. **Identify optimal designs** and generate predictions for experimental validation.

Mentoring outcomes: The student will develop skills in finite element analysis, scientific computing (Python), and data-driven optimization. They will present results at **ASME IDETC** or **UCUR** and contribute as co-author on a journal manuscript.

4 Student Research Project 2: Fabrication, CAD, and Experimental Testing

Student profile: One junior or senior undergraduate with coursework in machine design, CAD, or compliant mechanisms.

Scope and Deliverables

This student will lead fabrication and physical testing:

1. **Design tensegrity-inspired structures in CAD** (SolidWorks or Fusion 360) for multi-material 3D printing, with PLA struts and TPU tension elements printed on a single platform~~, incorporating interface design strategies (e.g. , interlocking geometries) to ensure durable PLA–TPU bonds [Ye et al., 2023]~~. Following Ye et al. [2023], we adopt a core-wrapping strategy where rigid PLA struts are encapsulated by continuous TPU skins, ensuring robust mechanical interlocking and preventing the interface delamination common in simple layered deposition.
2. **Fabricate structures** using the department's multi-material 3D printer, iterating on print parameters to achieve reliable tensegrity geometries.
3. **Conduct experimental testing:** compaction tests (quasi-static compression) ~~,~~ and drop tests (dynamic impact) ~~, and wave propagation measurements (hitting on one end and measuring time response on the other)~~following the drop-weight protocol of Pajunen et al. [2019] . Primary performance metrics are *specific energy absorption* (SEA) and *compaction efficiency* (ratio of absorbed energy to peak force)—tangible, operationally measurable outcomes suitable for undergraduate research. These experiments serve as the *high-fidelity* source.
4. **Compare experimental results** against simulation predictions from Student 1, quantifying model accuracy and informing surrogate updates.

Mentoring outcomes: The student will develop skills in CAD, additive manufacturing, test design, and data analysis. They will present at **UCUR** or **ASME IDETC** and contribute as co-author on a journal manuscript.

5 Mentoring Environment

The mentoring of undergraduate researchers is the central objective of this proposal. We are committed to providing a structured, supportive environment that develops students into independent, capable researchers.

Recruitment

Students will be recruited from courses in mechanics of materials, machine design, kinematics, and compliant mechanisms~~through brief in-class presentations~~. We target ~~junior or senior undergraduates. We will actively seek to recruit students~~ juniors/seniors and will actively recruit from underrepresented groups in engineering.

Mentoring Structure

- **Weekly individual meetings** (30 min) with ~~faculty PI (Prof~~Profs. Hill ~~) and Co-PI (Prof. Baird)~~ and Baird for technical guidance ~~, troubleshooting,~~ and professional development~~discussions~~.
- **Weekly lab group meetings** (1 ~~hour)including~~hr): progress updates, student ~~presentations on recent literature , career discussions, and~~ literature presentations, skill-building workshops (~~e.g.,~~ scientific writing, ~~conference~~ presentation skills, research ethics).
- **Graduate ~~student~~ co-mentor:** Prof. Baird's master's student ~~will serve as~~ provides day-to-day ~~co-mentor, providing hands-on~~ guidance in simulation, coding, ~~fabrication,~~ and experimental technique. This layered ~~mentoring model ensures students receive~~ model ensures timely support while the graduate student develops ~~their own~~ mentoring skills.
- **Progressive responsibility:** Students begin with structured tasks ~~(running simulations with provided scripts, assembling structures from detailed CAD plans)~~ and progress to independent design decisions, culminating in ownership of their research project.
- **Peer mentoring:** In Year 2, experienced students ~~will~~ mentor newly recruited ~~freshmen or sophomores, reinforcing their own learning while expanding the research group and~~ sophomores, creating a sustainable pipeline of trained researchers.

Student Development Outcomes

~~Students will develop a broad, transferable skill set spanning~~Technical risk-mitigation activities are structured as learning modules that develop transferable skills:

- **Technical:** experimental mechanics (test design, data acquisition, interface characterization), computation (FEA modeling, Python, Bayesian optimization), and design (CAD, multi-material 3D printing with wrapping strategies).
- **Communication:** lab notebooks, group presentations, conference talks, and co-authoring a peer-reviewed manuscript.
- **Professional:** project management, data-driven decision-making (interpreting Go/No-Go correlation results), collaboration with interdisciplinary teams, and preparation for graduate school or industry careers.

6 Expected Research Outcomes

1. **Trained undergraduate researchers:** 2–3 students with hands-on experience in advanced manufacturing, computational modeling, and data-driven optimization, prepared for graduate study or engineering careers.
2. **Conference presentations:** Student-led presentations at **UCUR** (Utah Conference on Undergraduate Research) and **ASME IDETC** (International Design Engineering Technical

Conferences).

3. **Journal submission:** One peer-reviewed manuscript submitted to the *ASME Journal of Mechanical Design* or *Smart Materials and Structures*, with undergraduate students as co-authors.
4. **Open-source tools:** Simulation code, optimization scripts, and experimental data published on GitHub for reproducibility.

7 Potential Impact of Work

This project advances ~~the design of architected materials~~ architected material design for energy absorption with ~~direct~~ applications in **protective equipment** (helmets, body armor), **packaging** (shock-absorbing inserts), and **aerospace** (lightweight impact-resistant structures). By demonstrating that multifidelity ~~Bayesian optimization~~ BO can efficiently navigate the tensegrity design space, we establish a generalizable framework ~~applicable to other classes of architected materials and metamaterials.~~

~~The interdisciplinary training—spanning structural mechanics, advanced manufacturing, and machine learning—prepares~~ for other metamaterial classes. The interdisciplinary training prepares students for graduate study or careers in ~~industries increasingly reliant on~~ data-driven design. ~~The open-source~~ Open-source dissemination of tools and data extends impact beyond BYU~~and enables other research groups to build on our work~~.

8 Timeline

Task	Y1 Fall	Y1 Winter	Y2 Fall	Y2 Winter
Student recruitment & onboarding	•			
Literature review	•	•		
Tensegrity parameterization & CAD	•	•		
Interface testing & print validation	•	•		
FEA simulation development		•	•	
LF–HF correlation gate (Go/No-Go)		•		
Multifidelity BO implementation		~~•~~	•	•
Multi-material 3D printing & fabrication		•	•	•
Impact testing & data collection			•	•
Simulation–experiment comparison				•
Conference presentations (UCUR/IDETC)			•	•
Journal manuscript preparation				•
Peer mentoring of new students			•	•

Table 1: Project timeline across four semesters.

9 Budget

Total requested: **$25,000** over 2 years. Per grant guidelines, no funds are allocated to equipment, tuition, or faculty wages.

Category	Description	Year 1	Year 2
Undergraduate wages	2–3 students, hourly RA	\$7,500	\$7,500
Graduate student wages	Partial RA for grad co-mentor	\$2,000	\$2,000
Supplies	Filament, load cells, DAQ, compute	\$2,750	\$750
Travel	Conference registration & travel	\$0	\$2,000
Other	Poster printing, consumables	\$250	\$250
Annual Total		**\$12,500**	**\$12,500**

Undergraduate wages (\$15,000): Supports 2–3 undergraduates for simulation, 3D printing, and testing. **Graduate student wages (\$4,000):** Prof. Baird's master's student serves as day-to-day co-mentor. **Supplies (\$3,500):** PLA/TPU filament, load cells, DAQ module, fasteners, and cloud compute credits (Year 1 heavy). **Travel (\$2,000):** Student presentations at ASME IDETC or UCUR (Year 2). **Other (\$500):** Poster printing, consumables, contingency.

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Biographical Sketches

Jeffrey R. Hill — Principal Investigator

Professional Preparation

Ph.D.	Mechanical Engineering	University of Utah
M.S.	Mechanical Engineering	Brigham Young University
B.S.	Mechanical Engineering	Brigham Young University

Appointments

- Associate Professor, Department of Mechanical Engineering, Brigham Young University (current)
- Research Engineer, Sandia National Laboratories (prior)

Products Most Closely Related to This Proposal

Publications and products related to smart materials, shape memory alloys, compliant mechanisms, and structural mechanics.

Synergistic Activities

- Mentoring of undergraduate and graduate students in experimental mechanics and smart materials research at BYU.
- Teaching courses in mechanics of materials, machine design, and kinematics in the Department of Mechanical Engineering.
- Collaboration with national laboratories on advanced materials characterization and structural design.

Sterling G. Baird — Co-Principal Investigator

Professional Preparation

Ph.D.	Materials Science and Engineering	University of Utah
B.S.	Materials Science and Engineering	Brigham Young University

Appointments

- Faculty, Department of Mechanical Engineering, Brigham Young University (current)

Products Most Closely Related to This Proposal

Publications and products related to materials informatics, Bayesian optimization, data-driven materials discovery, and machine learning for materials science.

Synergistic Activities

- Development of open-source tools for Bayesian optimization and materials informatics.
- Mentoring of graduate and undergraduate students in data-driven methods for materials science and engineering.
- Active contributions to the materials informatics community through publications, software, and educational resources.